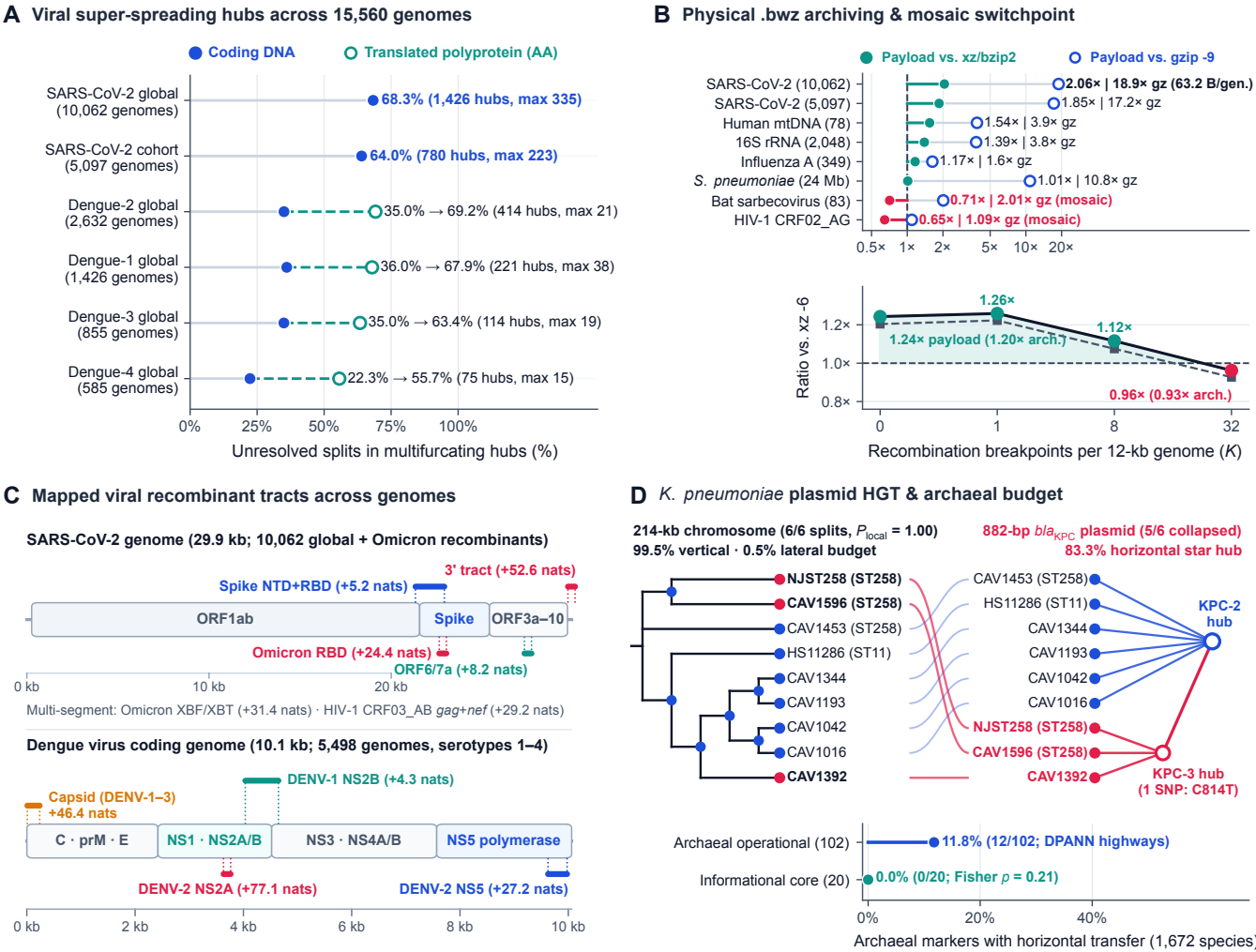

A Viral super-spreading hubs across 15,560 genomes
Coding DNA
Translated polyprotein (AA)
SARS-CoV-2 global (10,062 genomes)
68.3% (1,426 hubs, max 335)
SARS-CoV-2 cohort (5,097 genomes)
64.0% (780 hubs, max 223)
Dengue-2 global (2,632 genomes)
35.0% → 69.2% (414 hubs, max 21)
Dengue-1 global (1,426 genomes)
36.0% → 67.9% (221 hubs, max 38)
Dengue-3 global (855 genomes)
35.0% → 63.4% (114 hubs, max 19)
Dengue-4 global (585 genomes)
22.3% → 55.7% (75 hubs, max 15)
0% 25% 50% 75% 100%
Unresolved splits in multifurcating hubs (%)
B Physical .bwz archiving & mosaic switchpoint
Payload vs. xz/bzip2
Payload vs. gzip -9
SARS-CoV-2 (10,062)
2.06× | 18.9× gz (63.2 B/gen.)
SARS-CoV-2 (5,097)
1.85× | 17.2× gz
Human mtDNA (78)
1.54× | 3.9× gz
16S rRNA (2,048)
1.39× | 3.8× gz
Influenza A (349)
1.17× | 1.6× gz
S. pneumoniae (24 Mb)
1.01× | 10.8× gz
Bat sarbecovirus (83)
0.71× | 2.01× gz (mosaic)
HIV-1 CRF02_AG
0.65× | 1.09× gz (mosaic)
0.5× 1× 2× 5× 10× 20×
Ratio vs. xz -6
1.26×
1.12×
1.24× payload (1.20× arch.)
0.96× (0.93× arch.)
1.2× 1.0× 0.8×
0 1 8 32
Recombination breakpoints per 12-kb genome (K)
C Mapped viral recombinant tracts across genomes
SARS-CoV-2 genome (29.9 kb; 10,062 global + Omicron recombinants)
3' tract (+52.6 nats)
Spike NTD+RBD (+5.2 nats)
ORF1ab
Spike
ORF3a–10
Omicron RBD (+24.4 nats)
ORF6/7a (+8.2 nats)
0 kb 10 kb 20 kb
Multi-segment: Omicron XBF/XBT (+31.4 nats) · HIV-1 CRF03_AB gag+nef (+29.2 nats)
Dengue virus coding genome (10.1 kb; 5,498 genomes, serotypes 1–4)
DENV-1 NS2B (+4.3 nats)
Capsid (DENV-1–3) +46.4 nats
C · prM · E
NS1 · NS2A/B
NS3 · NS4A/B
NS5 polymerase
DENV-2 NS2A (+77.1 nats)
DENV-2 NS5 (+27.2 nats)
0 kb 2 kb 4 kb 6 kb 8 kb 10 kb
D K. pneumoniae plasmid HGT & archaeal budget
214-kb chromosome (6/6 splits, P local = 1.00)
99.5% vertical · 0.5% lateral budget
882-bp bla KPC plasmid (5/6 collapsed)
83.3% horizontal star hub
NJST258 (ST258)
CAV1596 (ST258)
CAV1453 (ST258)
HS11286 (ST11)
CAV1344
CAV1193
CAV1042
CAV1016
CAV1392
KPC-2 hub
KPC-3 hub (1 SNP: C814T)
Archaeal operational (102)
11.8% (12/102; DPANN highways)
Informational core (20)
0.0% (0/20; Fisher p = 0.21)
0% 20% 40%
Archaeal markers with horizontal transfer (1,672 species)